
Multimodal Explainable AI for Early-Stage Skin Cancer Classification

Using Dermoscopic Images and Clinical Metadata

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GitHub: <https://github.com/Ahmed-Mohsen-2005/XAI-skin-cancer-project>

Abstract

Early detection of melanoma is critical for patient survival, yet visual diagnosis remains a significant challenge due to morphological similarities between benign and malignant lesions. This paper presents a comprehensive multimodal Explainable AI (XAI) framework that integrates high-resolution dermoscopic images with structured clinical metadata — patient age, biological sex, and anatomical site — to improve both diagnostic accuracy and clinical transparency. We systematically evaluate sixteen distinct deep learning architectures spanning five fusion strategies: early token injection, gated fusion, bidirectional cross-attention, late fusion, and graph neural network fusion, using the ISIC-DICM-17K dataset of 17,060 perfectly balanced dermoscopic images paired with clinical metadata.

Our results demonstrate that deeper metadata integration consistently outperforms image-only and late-fusion baselines, with ViT Early Fusion achieving the highest performance (AUC: 0.9596, F1: 0.8915, Melanoma Recall: 91%). The EfficientNetV2-S Gated Fusion achieves the best AUC among late-fusion variants (0.9497), confirming that dynamic modality balancing outperforms static concatenation. To address the clinical "black-box" problem, we implement and compare six XAI techniques including Grad-CAM, Attention Rollout, SHAP, Integrated Gradients, Cross-Attention visualization, and Gate Analysis. This work establishes that metadata-conditioned visual reasoning, combined with dual-modality explainability, provides a viable and trustworthy pathway for clinically deployable AI-assisted skin cancer screening.

Keywords: melanoma detection, multimodal deep learning, explainable AI, dermoscopy, vision transformer, metadata fusion, Grad-CAM, SHAP, gated fusion, early fusion

1. Introduction

Melanoma is the most lethal form of skin cancer globally, yet it carries a near-100% survival rate when detected at Stage I [1]. The central clinical challenge is that early-stage melanoma is visually indistinguishable from common benign nevi, requiring expert dermatological evaluation that is not universally accessible. Automated deep learning systems have shown promising results in image-based classification, with state-of-the-art convolutional networks achieving accuracy comparable to board-certified dermatologists under controlled conditions [2]. However, two structural limitations persist across the majority of deployed systems.

First, current AI models are predominantly image-only systems, ignoring the rich clinical context — patient age, biological sex, anatomical lesion location, and medical history — that practicing dermatologists

routinely incorporate into diagnostic reasoning. Melanoma risk is known to increase substantially with age and varies by anatomical site, meaning a model that treats all images identically is fundamentally incomplete [3]. Second, and equally critical for clinical adoption, deep learning models operate as "black boxes": they output a probability score without any interpretable rationale. This opacity creates a liability barrier for physicians, who cannot ethically delegate diagnostic decisions to a system they cannot audit or explain to patients [4].

This research addresses both limitations through a unified multimodal XAI framework. We: (1) design and evaluate sixteen distinct architectures that fuse dermoscopic image features with structured clinical metadata across five fusion strategies; (2) implement six complementary XAI methods to generate both spatial and tabular explanations; and (3) identify which fusion design best balances accuracy, clinical recall, and interpretability. Our core hypothesis is that early integration of clinical metadata — allowing it to condition visual feature extraction from the first Transformer layer — will outperform post-hoc fusion strategies, and that attention-based architectures offer inherently more principled explainability than convolutional baselines requiring grafted-on visualization tools.

The study is structured around four research questions:

- RQ1: Do multimodal models integrating clinical metadata with dermoscopic images outperform image-only baselines for melanoma classification?
- RQ2: Which fusion strategy — early token injection, gated, cross-attention, or late concatenation — produces the best classification performance?
- RQ3: Do XAI techniques produce clinically meaningful explanations consistent with established dermatological knowledge?
- RQ4: Is there a measurable performance-interpretability tradeoff across model architectures?

2. Review of Relevant Literature

2.1 Image-Only Classification Baselines

The earliest high-performing approaches to automated skin lesion classification relied exclusively on convolutional neural networks (CNNs) applied to dermoscopic images. Kassem et al. achieved 94.92% accuracy on an eight-class ISIC 2019 task using a fine-tuned GoogleNet with transfer learning [5]. Ahmad et al. similarly trained a custom CNN on 20,000 dermoscopic images, reaching over 90% accuracy and an AUC of approximately 0.95 [6]. Mahmud et al. advanced this paradigm architecturally by enhancing the Xception backbone with Global Average Pooling, Batch Normalization, and Swish activations, achieving 96.48% accuracy across two independent datasets [7]. Despite strong benchmark performance, all three studies share a structural limitation: they treat the dermoscopic image as the sole diagnostic input, ignoring patient-level clinical variables that a practicing dermatologist would routinely consider.

2.2 Explainability Methods

Explainability in skin lesion AI has evolved from purely post-hoc visualization tools toward architecturally integrated mechanisms. Grad-CAM and its variants represent the dominant approach: by computing gradients of the predicted class score with respect to convolutional layer activations, they produce spatial heatmaps highlighting image regions that influenced each prediction [7]. Nigar et al. demonstrated that integrating LIME into ResNet-18 improved clinician trust without reducing performance (94.47% accuracy, ISIC 2019) [8]. Muduli et al. integrated Grad-CAM into a custom attention-gated U-Net achieving 92.75% classification accuracy, with heatmaps confirming model attention to lesion boundaries rather than surrounding skin [9]. Gamage et al. employed Grad-CAM++ within a U2-Net mask-guided pipeline, achieving 98.37% accuracy with a web application receiving an 86.87% usability score from

medical experts [10]. Ruga et al. demonstrated the value of dual-modality XAI by combining Grad-CAM for spatial attribution with SHAP for tabular feature importance within their MultiExCam ensemble, achieving AUC scores of 91–98% across datasets [11]. Hong and Lin applied LIME diagnostically, revealing that misclassified cases frequently corresponded to instances where the model attended to irrelevant background features rather than the lesion itself [12]. Raju et al. introduced XAI-SkinCADx, a six-stage pipeline combining BiLSTM, deep ensemble CNNs, and Grad-CAM++ to achieve 95.63% accuracy and AUC 0.97, though the system's complexity limits real-time deployment [13].

2.3 Multimodal and Clinical Context Approaches

The most clinically faithful AI systems incorporate both image and patient metadata. Ahammed et al. constructed the ISIC-DICM-17K dataset — 17,060 balanced dermoscopic images paired with age, sex, and anatomical site — and demonstrated that multimodal fusion improved discrimination between melanoma and non-melanoma over image-only baselines, achieving AUC 0.8851 [1]. Aksoy et al. deployed XRAI explainability within a complete web-based clinical decision support platform integrating EfficientNet-B3 with patient metadata, achieving 89.09% on HAM10000 with a four-tier risk warning system [14]. Metta et al. proposed an interpretable framework using the ABELE method combining ResNet with an adversarial autoencoder, generating exemplar and counter-exemplar explanations that significantly increased clinician trust, particularly among non-experts [15]. Both multimodal studies confirm that metadata integration is clinically meaningful, while identifying unresolved challenges: missing metadata fields, fusion architecture design choices, and the absence of rigorous external validation across diverse patient populations. Our work directly addresses these gaps through systematic multi-strategy evaluation and comprehensive XAI analysis.

3. Methodology

3.1 Dataset and Preprocessing

All experiments used the ISIC-DICM-17K dataset, comprising 17,060 dermoscopic images with a perfectly balanced class distribution (8,530 melanoma, 8,530 non-melanoma) [1]. Table 1 summarizes key dataset properties.

Table 1: ISIC-DICM-17K Dataset Properties

Property	Details
Dataset	ISIC-DICM-17K (International Skin Imaging Collaboration)
Total Samples	17,060 dermoscopic images + clinical metadata CSV
Class Balance	8,530 Melanoma 8,530 Non-Melanoma (perfectly balanced)
Image Format	JPEG; resized to 224×224 (ViT/CNN) or 384×384 (EfficientNetV2-S)
Metadata Features	Age, Sex, Anatomical Site, History of MM, Dermoscopic Type, Diagnosis Confirm Type → 29 encoded features
Split Strategy	Stratified 70/15/15 train/val/test; 5-fold CV for ViT-based models
Normalization	ImageNet statistics: mean=[0.485, 0.456, 0.406], std=[0.229, 0.224, 0.225]
Imputation	Median for age (continuous); mode for categorical features; MinMax scaling for age
Source	Ahammed et al. (2025); curated at Manchester Metropolitan University

Clinical metadata features include patient age (continuous, median-imputed for missing values), biological sex (one-hot encoded), anatomical site (one-hot encoded), history of melanoma, dermoscopic type, and diagnosis confirmation type — yielding 29 total encoded clinical features. A stratified 70/15/15 train/validation/test split was applied globally prior to any training, with five-fold stratified cross-validation used for ViT-based model selection.

3.2 Augmentation Strategy

Training augmentation was restricted to transforms that preserve medically relevant visual features. Table 2 details the augmentation strategy, including explicit exclusions.

Table 2: Data Augmentation Strategy

Transform	Parameter	Rationale
HorizontalFlip	p=0.5	Lesions are anatomically symmetric
VerticalFlip	p=0.5	Orientation-invariant in dermoscopy
RandomRotate90	p=0.5	Captures lesion from multiple orientations
Rotate $\pm 15^\circ$	p=0.5	Mild rotation preserves morphological cues
ShiftScaleRotate	shift/scale ± 0.05 , p=0.4	Simulates minor device repositioning
RandomBrightnessContrast	limit=0.10, p=0.4	Simulates inter-device lighting variation
CLAHE	clip=2.0, p=0.2	Enhances pigment network; low p preserves diagnostic color
CutMix + MixUp	p=0.30 / p=0.20	Applied in EfficientNetV2-S models for regularization
Excluded: Grayscale, aggressive ColorJitter, large CoarseDropout	—	Color and spatial integrity are diagnostically critical in dermoscopy

Validation and test transforms were restricted to resize and ImageNet normalization only. This constraint was strictly enforced to prevent any data leakage through test-time statistics.

3.3 Model Architectures and Fusion Strategies

Sixteen architectures were implemented across five fusion paradigms. All models share common training infrastructure: AdamW optimization with weight decay $1e-4$, differential learning rates (backbone: 2×10^{-5} , heads: 2×10^{-4}), OneCycleLR scheduler with 10% linear warmup, Focal Loss ($\alpha=0.25$, $\gamma=2.0$) with label smoothing ($\epsilon=0.10$), gradient clipping (max_norm=1.0), AMP mixed-precision training, gradient accumulation over two steps (effective batch size 32), and progressive backbone unfreezing.

3.3.1 Early Token Injection (ViT Early Fusion)

The ViT Early Fusion model (Jana Ahmed) uses ViT-Base/16 pretrained on ImageNet. Clinical metadata is projected through a two-layer MLP (Linear→LayerNorm→GELU→Dropout→Linear→LayerNorm) from the 29-feature metadata dimension to 768-dim, then injected as a learnable [META] token into the patch sequence before the first Transformer block. The 198-token sequence ([CLS, P1...P196, META]) passes through all twelve Transformer layers, allowing metadata to condition visual feature extraction at every level of the hierarchy. The classifier reads the [CLS] token through a LayerNorm→Linear(768→256)→GELU→Dropout(0.3)→Linear(256→2) head. The Hybrid ViT-MLP

variant shares the same backbone but processes metadata independently, merging at the classifier head only — enabling direct comparison between early and late fusion with identical backbones.

3.3.2 Gated Fusion

Gated fusion architectures use a learned sigmoid gate $g = \sigma(W \cdot [\text{img_feat}, \text{meta_feat}])$ to dynamically weight each modality per sample: $\text{fused} = g \odot \text{img} + (1 - g) \odot \text{meta}$. This design allows the model to adapt to variable input quality — increasing reliance on metadata when image quality is degraded by hair, gel artifacts, or poor focus. Three gated models were implemented: EfficientNetV2-S + TabTransformer at 384×384 (Abdelrahman Mohamed), MaxViT-T with multi-scale attention (Ahmed Mohsen), and DaViT-Tiny with dual channel and spatial attention (Ahmed Mohsen). EfficientNetV2-S additionally applied CutMix (p=0.30) and MixUp (p=0.20), and used 5-step Test-Time Augmentation at inference.

3.3.3 Bidirectional Cross-Attention

Cross-attention architectures model asymmetric inter-modal dependencies. Image patch tokens serve as queries while metadata tokens serve as keys and values, asking: "given this patient's age and anatomical site, which visual features are most suspicious?" Three cross-attention models were implemented: EfficientNet-B3 with standard multi-head cross-attention (Abdelrahman Mohamed), DINOv2-ViT-S/16 + TabTransformer with bidirectional cross-attention over two alternating layers (Ahmed Mohsen), and BioViL-T + FT-Transformer with lesion-conditioned sparse attention (topK=8) using a medically pretrained image encoder (Ahmed Mohsen). Additional cross-attention variants include EfficientNetV2-S + TabTransformer + bidirectional cross-attention (Jana Ahmed) and Swin Transformer V2 + TabTransformer + cross-attention (Jana/Abdelrahman).

3.3.4 Late Fusion and Specialized Architectures

The EfficientNet-B3 Late Fusion baseline (Malak Mohamed) concatenates independent EfficientNet-B3 image features (1536-dim) with a two-layer MLP metadata branch (64-dim) before a final classifier head, serving as a stable lower-bound reference. The ConvNeXt + FT-Transformer + Bilinear Fusion model replaces concatenation with multiplicative bilinear interactions to capture higher-order cross-modal dependencies. The GNN + MaxViT-T model (Malak Mohamed) constructs a graph over MaxViT-T patch embeddings with metadata as node attributes, using Graph Attention Networks to aggregate spatial co-occurrence patterns in lesion morphology. Additional architectures include ViT Early Fusion, Hybrid ViT-MLP (Malak Mohamed), and a CLIP-style Contrastive Multimodal Model (Abdelrahman Mohamed).

3.4 Explainability Methods

Six complementary XAI techniques were implemented across all models to provide spatial, attribution-based, attention-based, perturbation-based, and gate-level explanations. Table 3 summarizes each method.

Table 3: XAI Methods Summary

XAI Method	Implementation	Clinical Interpretation
Grad-CAM	Backward hook on final conv/feature stage; spatial maps weighted by malignant-class gradient; token grids reshaped to 2D for ViT/DaViT	Identifies which image regions drove the malignant prediction
Attention Rollout	Recursive layer-by-layer multiplication of attention matrices across all 12 ViT blocks; residual-aware weighting	Produces sharper spatial localization than single-layer attention; suitable for multi-layer architectures
SHAP KernelExplainer	50-sample kmeans background; 200 perturbations; applied to metadata branch; beeswarm + bar + waterfall plots	Quantifies importance of age, sex, anatomical site and other clinical features per prediction

Integrated Gradients	Captum; 50 interpolation steps from zero baseline; pixel-level and tabular attribution	Theoretically grounded attribution; confirmed 80%+ overlap with annotated lesion boundaries
Cross-Attention Maps	Raw attention weight matrices from cross-attention layers; visualizes clinical token → image patch relationships	Reveals which clinical features (e.g., age) attend to specific lesion regions
Gate Analysis	Gate activation $g = \sigma(W \cdot [\text{img}, \text{meta}])$ logged per sample; distribution plotted over test set	Confirms dynamic modality balancing: low-quality images shift weight toward metadata

4. Results

4.1 Overall Model Performance

Table 4 summarizes validation performance across all implemented architectures. Five-fold stratified cross-validation was used for ViT-based models; stratified single splits for remaining architectures. All metrics are from the best validation fold prior to held-out test evaluation.

Table 4: Complete Model Performance Summary (ISIC-DICM-17K Validation Set)

Model	Member	Val Acc	F1	AUC-ROC	Mel. Recall	XAI
ViT Early Fusion	Jana	89.15%	0.8915	0.9596	91%	Attn Rollout
EfficientNetV2-S Gated	Abdelrahman	87.22%	0.8720	0.9497	91%	Grad-CAM + Gate
Hybrid ViT-MLP	Jana	86.96%	0.8695	0.9493	89%	Grad-CAM + SHAP
EfficientNet-B3 Late Fusion	Malak	85.53%	0.8553	~0.94	86%	Grad-CAM
EfficientNet-B3 CrossAttn	Abdelrahman	84.16%	0.7900	0.9262	87%	Grad-CAM
ConvNeXt + FT-Transformer	Abdelrahman	82.88%	~0.83	~0.91	83%	Grad-CAM
MaxViT-T + Gated Fusion	Ahmed	~82%	~0.82	0.9429	85%	Grad-CAM + SHAP
GNN + MaxViT-T Fusion	Malak	83.86%	0.8384	0.9265	84%	Grad-CAM + GNN
ConvNeXt-Tiny + Bidir. CrossAttn	Ahmed	81.21%	~0.81	0.8989	94%*	Grad-CAM
DaViT-Tiny + Gated Fusion	Ahmed	~81.86%	~0.819	0.9528	84%	Grad-CAM + SHAP
DenseNet121 + Swish MLP	Abdelrahman	79.95%	~0.76	0.8957	80%	Grad-CAM
BioViL-T + FT-Transformer + Sparse Attn	Ahmed	—	—	—	—	Grad-CAM + IG
DINOv2-ViT-S/16 + TabTransformer CrossAttn	Ahmed	—	—	—	—	CrossAttn Maps

ViT Early Fusion + Swin V2 + SegFormer-B5 (Jana)	Jana	—	—	—	—	Attn + SHAP
CLIP-style Contrastive Model (Abdelrahman)	Abdelrahman	—	—	—	—	SHAP + IG

* ConvNeXt-Tiny sensitivity reflects a sensitivity-specificity tradeoff caused by missing ImageNet normalization during training.

ViT Early Fusion achieved the highest overall performance (Accuracy: 89.15%, F1: 0.8915, AUC: 0.9596), with mean cross-validated F1 of 0.8864 ± 0.0029 — confirming robust generalization across data splits. EfficientNetV2-S Gated Fusion achieved the highest AUC among late-fusion architectures (0.9497) with 91% melanoma recall. DaViT-Tiny achieved AUC 0.9528, the highest among Ahmed's models, indicating strong ranking ability despite lower overall accuracy. The EfficientNet-B3 Late Fusion baseline demonstrated the highest cross-fold stability (std = 0.0020), confirming its role as a reliable lower bound. Test-Time Augmentation provided consistent +0.3–0.5% AUC gains across all EfficientNetV2-S models.

4.2 ViT Early Fusion — Best Model Detail

Table 5 presents the per-class classification report for the best fold (Fold 5) of the ViT Early Fusion model, which represents the single best configuration across the entire study.

Table 5: ViT Early Fusion — Best Fold (Fold 5) Classification Report

Class / Average	Val Acc	F1	AUC-ROC	Precision	Recall
Non-Melanoma	88%	0.87	0.9596	0.90	0.88
Melanoma	89%	0.89	0.9596	0.88	0.91
Macro Average	89.15%	0.8915	0.9596	0.89	0.89

The model achieves 91% melanoma recall, meaning only 9% of melanoma cases are missed — a clinically critical metric for screening applications where false negatives carry the highest cost. Cross-validation variance was extremely low (F1 range: 0.8836–0.8915, std=0.0029), confirming robust generalization. The model's 5-fold results were: Fold 1: 0.8836, Fold 2: 0.8851, Fold 3: 0.8844, Fold 4: 0.8874, Fold 5 (Best): 0.8915.

4.3 Fusion Strategy Comparison

Table 6 presents performance by fusion strategy, using the best-performing model per strategy. The hierarchy is unambiguous: deeper integration consistently outperforms post-hoc fusion.

Table 6: Fusion Strategy Performance Comparison

Fusion Strategy	Best Model	F1	AUC	Key Property
Early Token Injection	ViT Early Fusion	0.8915	0.9596	Metadata in all 12 Transformer layers
Gated Fusion	EfficientNetV2-S Gated	0.8720	0.9497	Dynamic per-sample modality weighting
Late Fusion (Concat)	EfficientNet-B3 Late Fusion	0.8553	~0.94	Independent branches, most stable (std=0.002)

Bilinear/GNN Fusion	ConvNeXt + FT-Transformer	~0.83	~0.91	Multiplicative cross-modal dependencies
Bidirectional Cross-Attention	EfficientNet-B3 CrossAttn	0.79	0.9262	High expressiveness; sensitive to dataset scale

4.4 XAI Analysis

Attention Rollout maps for ViT Early Fusion consistently highlight the central lesion region, irregular borders, and asymmetric pigmentation — features that align precisely with the ABCD criteria (Asymmetry, Border, Color, Diameter) used in clinical dermoscopy. For melanoma cases, rollout maps concentrate on dark irregular border regions and satellite lesions; for non-melanoma cases, attention distributes more uniformly across symmetric, well-bounded structures. The rollout approach produced sharper localization than single-layer attention extraction across all tested folds.

Grad-CAM heatmaps across CNN-based and hybrid models (EfficientNet-B3, DaViT, MaxViT, ConvNeXt) confirm consistent localization of diagnostically relevant features. EfficientNetV2-S Gated Fusion heatmaps highlight lesion centers, irregular borders, dark asymmetric structures, and internal pigmentation changes. Gate activation analysis confirmed the predicted dynamic modality balancing behavior: high-quality images produced gate values $g > 0.7$ (high image reliance), while blurred or artifact-heavy images shifted weight toward clinical metadata ($g < 0.4$).

SHAP analysis of the metadata branch identified age and anatomical site as the two most influential clinical features across all models where applied. This aligns with established melanoma epidemiology — older patients (>60) and posterior torso/acral sites carry substantially elevated risk. Beeswarm plots from DaViT additionally reveal that melanocytic type and dermoscopic type contribute non-trivially to high-confidence melanoma predictions. Integrated Gradients confirmed that attributed pixels overlap with manually annotated lesion boundaries in over 80% of test cases, providing strong evidence of clinically valid visual attention.

5. Discussion

The results establish a clear performance hierarchy organized by fusion depth: early integration $>$ gated fusion $>$ bilinear/GNN fusion $>$ bidirectional cross-attention $>$ late concatenation. Early Fusion outperforms all other strategies because metadata tokens participate in all 12 Transformer self-attention layers, enabling the model to learn conditioned visual representations — the model "sees" a lesion differently depending on whether the patient is 25 or 75 years old, or whether the lesion site is the palm versus the trunk. This conditioning is fundamentally unavailable to late-fusion designs, where both branches process their respective inputs independently before a final concatenation step.

Gated Fusion represents the optimal late-fusion design: its dynamic per-sample modality weighting (AUC 0.9497) outperforms static concatenation (AUC ~ 0.94) by adapting to variable-quality inputs. When dermoscopy image quality is degraded by hair, gel artifacts, or poor focus, the gate appropriately increases reliance on metadata. This behavior was empirically confirmed through gate activation analysis, supporting the use of gated architectures in real-world deployment settings where image quality cannot be controlled.

Bidirectional Cross-Attention models showed strong theoretical motivation but underperformed relative to Early Fusion, primarily due to overfitting on the 17,060-sample dataset. The ConvNeXt bidirectional model demonstrated a sensitivity of 94.1% alongside a specificity of only 68.5% — an uncalibrated decision threshold compounded by the missing ImageNet normalization identified as a known implementation error. The FT-Transformer bilinear fusion model produced the largest train-validation gap

across all experiments, confirming that model complexity must be matched by dataset scale to avoid memorization rather than generalization.

Across all architectures, SHAP consistently identified age as the single most influential clinical variable, with older patients (>60) receiving substantially higher malignancy risk estimates for morphologically borderline lesions. This finding aligns with clinical knowledge and provides an auditable, patient-specific explanation that complements the spatial heatmaps produced by Grad-CAM. The dual-modality explanation — visual (where) and tabular (why) — mirrors the two-channel information processing of a practicing dermatologist, making the system's reasoning interpretable along both diagnostic axes.

Comparing with prior work: our ViT Early Fusion (AUC 0.9596) exceeds the best multimodal baseline reported by Ahammed et al. [1] on the same dataset (AUC 0.8851) — a +7.4% improvement attributable to the deeper early-fusion integration strategy and the modern ViT-Base/16 backbone. Our models also outperform the image-only baselines of Ahmad et al. [6] (AUC ~0.95) and Aksoy et al. [14] (89.09% accuracy on HAM10000), providing empirical validation that metadata integration adds measurable diagnostic value.

Several limitations require acknowledgment. The SHAP baseline used zero-image tensors, likely understating true joint-modality metadata contributions; future work should use mean-image baselines. The 4% train-validation gap observed in ViT Early Fusion indicates residual overfitting addressable through stochastic depth and stronger regularization. External validation on diverse clinical cohorts, particularly across different Fitzpatrick skin phototypes, remains necessary before deployment.

6. Ethical and Legal Analysis

The deployment of AI systems in clinical diagnostic workflows raises substantial ethical, legal, and epistemic concerns that cannot be subordinated to accuracy benchmarks. Three dimensions are directly relevant to this research.

Under the European Union General Data Protection Regulation (GDPR) Article 22 and the EU AI Act, systems performing automated medical decision-making must provide meaningful explanations to affected individuals. Our implementation of Grad-CAM, Attention Rollout, and SHAP directly addresses the "Right to Explanation" by generating both visual and tabular justifications for each prediction. The system is designed as a Decision Support Tool rather than an autonomous diagnostic agent: it provides ranked evidence for clinician review, preserving human final authority over diagnostic conclusions. This design distinction is legally significant under the EU AI Act's risk classification framework, which categorizes autonomous medical diagnosis AI as high-risk and subject to conformity assessment requirements.

Algorithmic bias represents a second dimension of ethical concern. The ISIC-DICM-17K dataset is class-balanced (8,530 melanoma vs. 8,530 non-melanoma), reducing the risk of accuracy-driven bias toward the majority class. However, the dataset's demographic composition — predominantly lighter Fitzpatrick skin types — means that model performance on darker skin phototypes remains unvalidated. Melanoma presents differently across phototypes, and a system that performs at AUC 0.96 on the training distribution may perform substantially worse on underrepresented populations. We flag this as a critical limitation requiring prospective validation across diverse ethnographic cohorts before any deployment, in accordance with the principle of Distributive Justice.

Third, clinical liability under current legal frameworks remains unresolved. A clinician who acts on an AI recommendation that results in patient harm may face liability questions depending on whether the AI was used as a certified medical device or an advisory tool. By providing transparent XAI outputs, our system enables clinicians to exercise informed judgment rather than opaque delegation, reducing but not

eliminating this exposure. Future work should engage directly with regulatory bodies to establish evidence standards for AI-assisted dermatological diagnosis.

7. Conclusion

This research presents a systematic evaluation of sixteen multimodal deep learning architectures for skin lesion classification, demonstrating that the depth of metadata integration is the primary determinant of performance. ViT Early Fusion, by conditioning all twelve Transformer layers on clinical context, achieved the highest results (AUC: 0.9596, F1: 0.8915, Melanoma Recall: 91%), establishing early fusion as the preferred strategy when computational resources permit. Gated Fusion provides the optimal late-fusion alternative through dynamic per-sample modality balancing (AUC: 0.9497), with gate analysis confirming empirically meaningful behavior rather than static preferences.

The importance of explainability is validated at multiple levels. Attention Rollout maps align with ABCD dermatological criteria. SHAP attribution identifies age and anatomical site as the two most clinically predictive metadata features. Grad-CAM heatmaps consistently localize diagnostically relevant visual structures. Integrated Gradients confirm that attributed pixels overlap with annotated lesion boundaries in over 80% of cases. Together, these XAI methods transform our models from black-box classifiers into transparent reasoning systems that clinicians can interrogate, audit, and selectively trust.

Our work establishes that performance and interpretability are not inherently in tension for architectures designed around genuine clinical reasoning. Future directions include: (1) external validation across diverse skin phototypes and clinical cohorts; (2) replacement of zero-image SHAP baselines with clinically representative references; (3) application of mean-image Integrated Gradients for full multimodal attribution; (4) prospective clinical evaluation with dermatologist user studies; and (5) extension to multi-class lesion classification beyond the binary melanoma/non-melanoma task.

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